MyBudget

A personal budget app for Android, built with Flutter. It records income and expenses, sorts them into categories, sets spending limits and shows where the money goes. Everything is stored on the phone; a few free online services add avatars, icons, exchange rates and holiday dates.

1. Users module	Developer 1
2. Categories module	Developer 2
3. Transactions module	Developer 3
4. Budgets module	Developer 4
5. Dashboard and statistics	Shared

Overview

How the app works

MyBudget is offline first. All data lives in a SQLite database on the phone, and every screen reads it through four modules that each own one table. The online services are optional: when there is no network the app shows a fallback and keeps working.

A typical session

1. Choose a profile
Create one, or pick an existing profile. Its avatar, currency and theme follow it.

2. Organise
Thirteen categories are ready; add your own with any icon.

3. Record
Add income and expenses by hand, in another currency, or by scanning a receipt.

4. Control
Set budgets, watch their state, and check the dashboard and statistics.

Navigation

- The bottom bar opens Dashboard, Transactions, Budgets and More.
- More leads to Categories, Profile and Statistics.
- Lists slide in row by row, screens fade between each other, and loading shows grey skeleton rows. All motion is turned off when the phone asks for reduced motion.

Modules

Module	Owns	Main screens
Users	Profiles	Profile gate, profile, edit profile
Categories	Categories	Category list, category form, category picker
Transactions	Transactions	History, transaction form, receipt scan
Budgets	Budgets	Budget list, budget form
Shared	Nothing (reads all)	Dashboard, statistics

Free APIs

Service	Module	What it adds
DiceBear avatars api.dicebear.com	Users	Draws a unique avatar for each profile. The seed is the profile id and creation time, never the name or email, so no personal data leaves the phone. Offline, the profile's initial is shown.
Iconify api.iconify.design	Categories	Icon search in the category form. Only Material Design Icons, Tabler and Material Symbols are searched, all free to use (Apache 2.0 or MIT). A chosen icon is downloaded once and kept on the phone, so it still shows offline.

ExchangeRate-API open.er-api.com	Transactions	Daily exchange rates for the “Paid in” conversion, including TND. The rate is cached for 12 hours and credited on screen as its terms require.
Google ML Kit text recognition on the device, no server	Transactions	Reads the text of a receipt photo for the scanner. It runs on the phone, works offline and the photo is never uploaded.
Nager.Date date.nager.at	Budgets	Tunisian public holidays such as Independence Day and Republic Day.
AlAdhan api.aladhan.com	Budgets	The Hijri calendar, used to find Ramadan, Eid al-Fitr, Eid al-Adha, the Islamic New Year and Mawlid. Both calendar answers are cached for a month.

Answers from the online services are cached on the phone, so the app keeps the last result offline. None of them needs an account or a key.

Module 1 · Developer 1

Users

The users module manages the profiles stored on the device. A profile owns every other record in MyBudget: its categories, transactions and budgets. There is no sign-in; the email only identifies the profile. Each profile gets a generated avatar and can switch the whole app between dark and light.

What it does

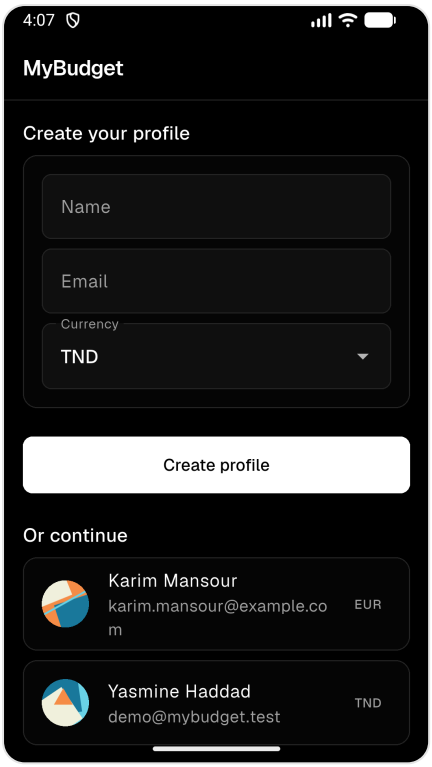
- Opens on profile creation when the device has no profile, and on profile choice when it has several.
- Creates a profile with a name, a unique email and a currency (TND, EUR, USD, GBP or MAD).
- Validates the form and explains each error under its field.
- Shows the active profile with its avatar, currency and creation date.
- Edits the name, email and currency; switches between profiles; deletes a profile with all its data.
- Switches the app between the dark theme (default) and the light theme; the choice is remembered.
- Asks the categories module to add the predefined categories to every new profile.

Free APIs used

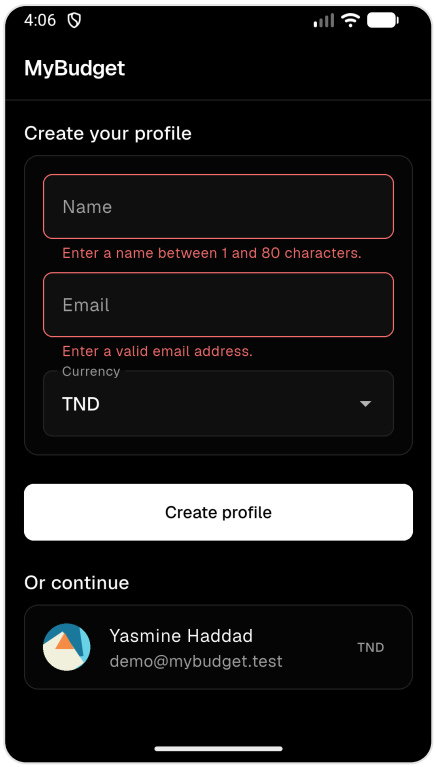
DiceBear avatars api.dicebear.com

Draws a unique avatar for each profile. The seed is the profile id and creation time, never the name or email, so no personal data leaves the phone. Offline, the profile's initial is shown.

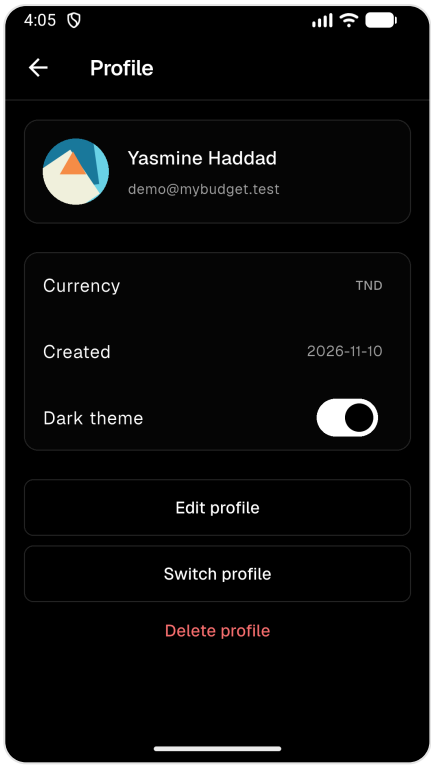
Screens



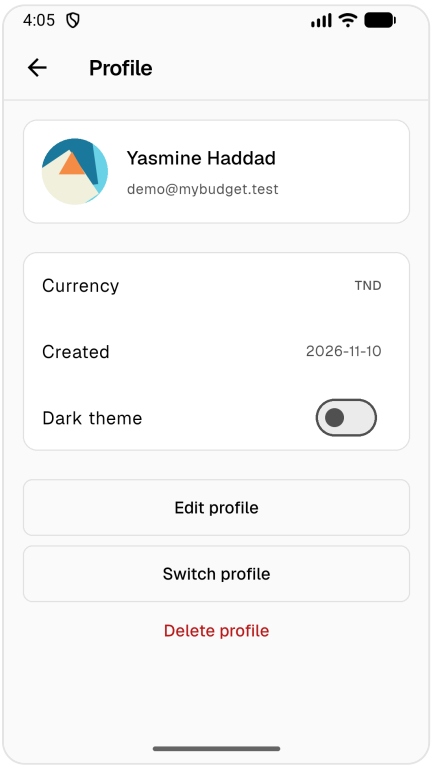
Profile gate
Create a profile, or continue with one already on the device. Each profile shows its avatar and currency.



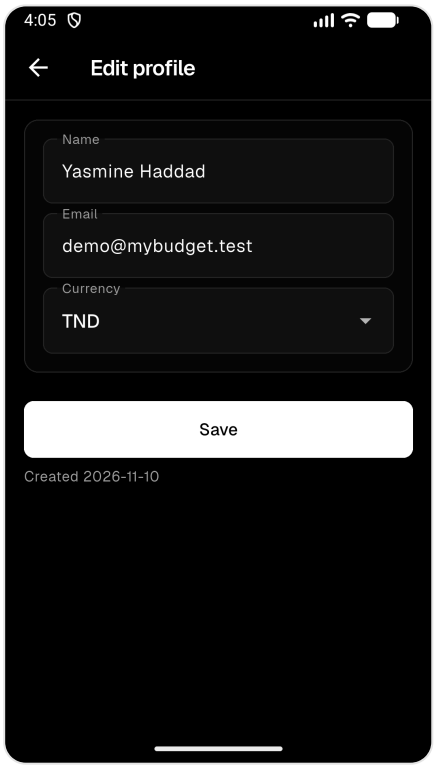
Form validation
An empty name and an invalid email are refused, with a message under each field.



Active profile
Avatar, details and the dark theme switch, with edit, switch and delete actions.



Light theme
The same screen after turning the dark theme off.



Edit profile
Name, email and currency can be changed. A new currency changes the label on amounts.

Module 2 · Developer 2

Categories

The categories module classifies money. Every category belongs to one profile and is either an expense or an income category. Thirteen predefined categories are added to each new profile, and the owner can add custom ones with an icon from the built-in set or from thousands of free online icons. The other modules choose categories through the picker this module provides.

What it does

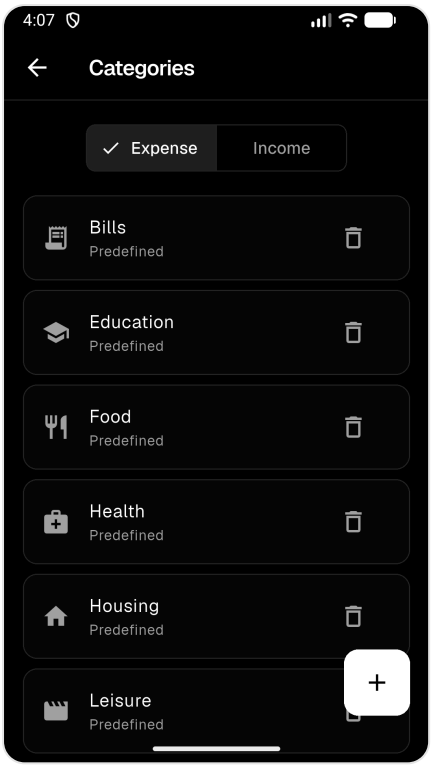
- Lists the profile's categories, split into expense and income, marked Predefined or Custom.
- Creates a category with a name, a type and an icon from the 20 built-in icons.
- Searches free online icons by keyword and uses the chosen one like a built-in icon.
- Edits a category's name and icon; refuses a type change while the category is in use.
- Refuses duplicate names of the same type (case-insensitive).
- Deletes an unused category. A used one is kept until its transactions and budgets are moved to another category of the same type, which the screen offers.
- Provides the category picker used by the transaction and budget forms, filtered by type.

Free APIs used

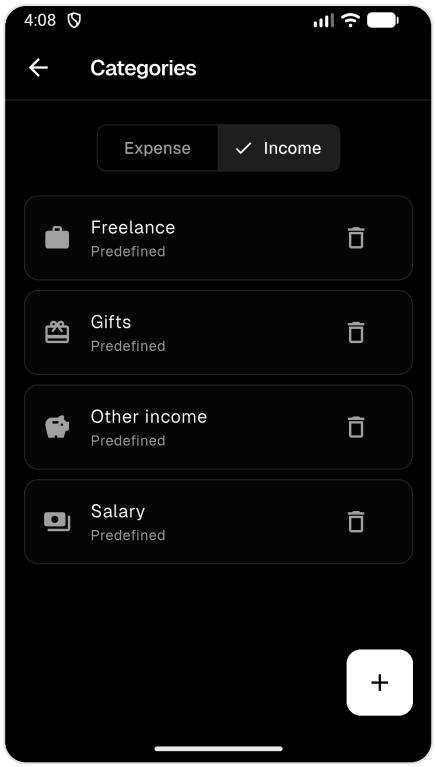
Iconify api.iconify.design

Icon search in the category form. Only Material Design Icons, Tabler and Material Symbols are searched, all free to use (Apache 2.0 or MIT). A chosen icon is downloaded once and kept on the phone, so it still shows offline.

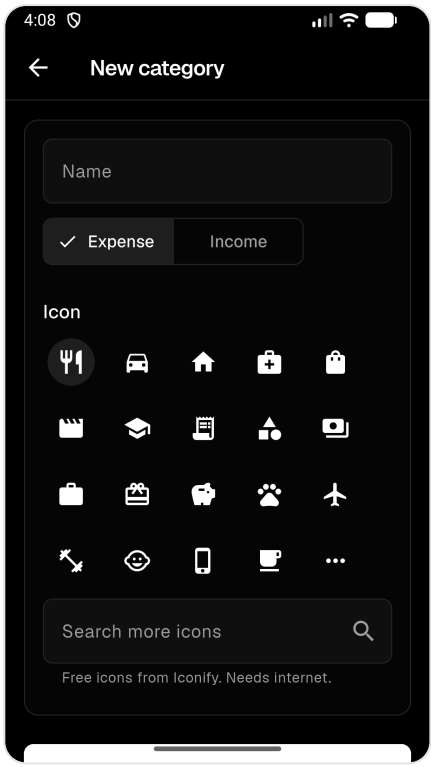
Screens



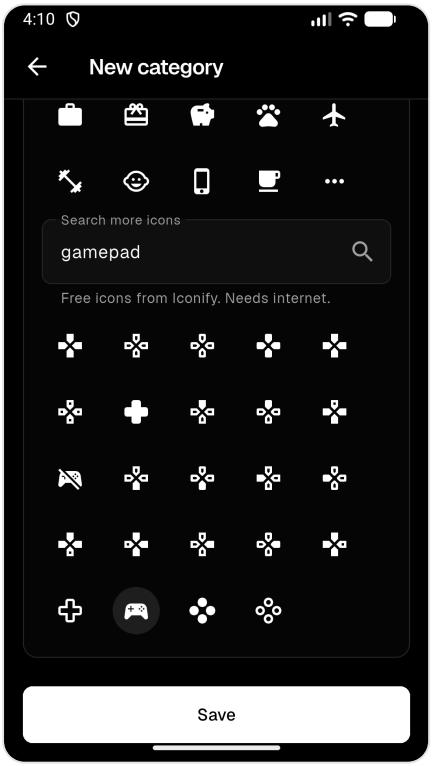
Expense categories
The profile's expense categories, predefined and custom.



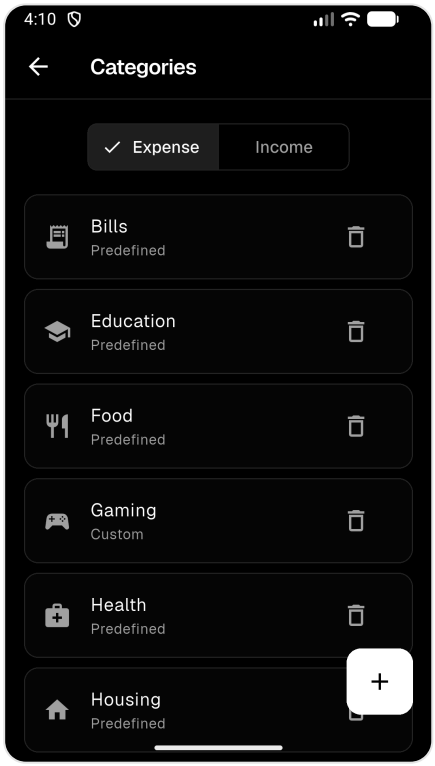
Income categories
The same list switched to income.



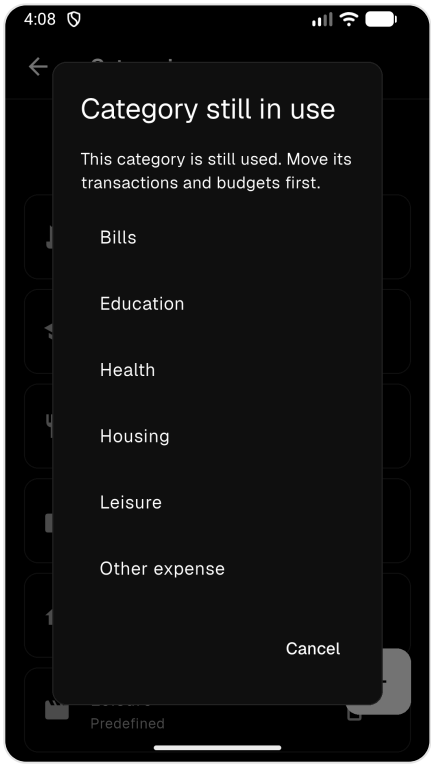
New category
Name, type, the 20 built-in icons and the online icon search.



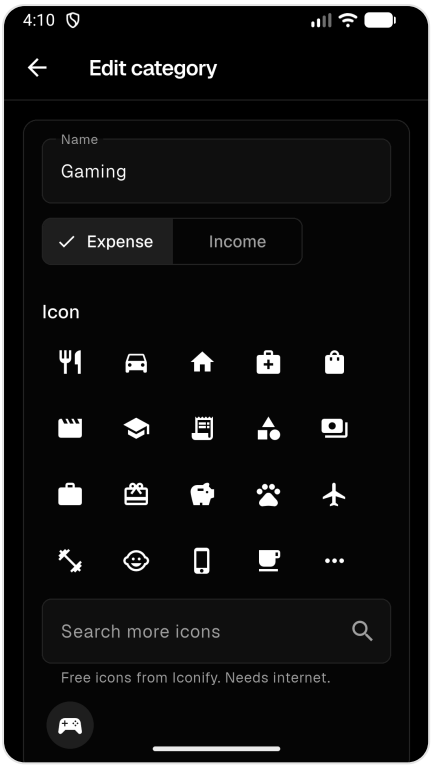
Online icon search
Searching “gamepad” returns free icons from Iconify; the controller is selected.



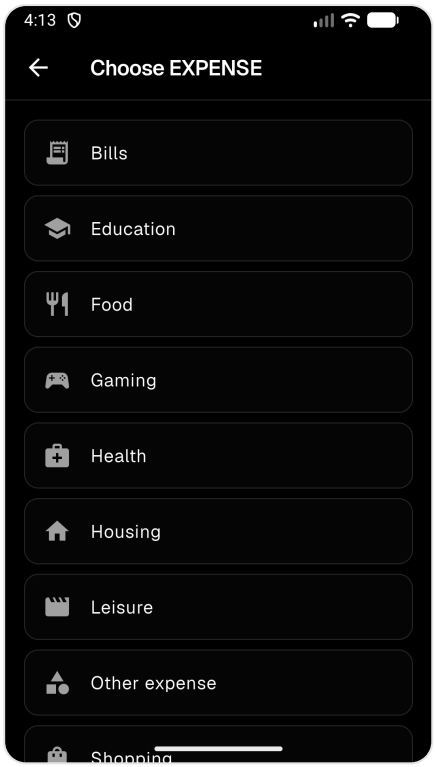
Custom category
The new Gaming category in the list, with its downloaded icon.



Category still in use
Food has transactions and a budget, so the app offers to move them first.



Edit category
The Gaming category opened for editing, keeping its online icon.



Category picker
The shared picker used by the other modules, here limited to expenses.

Module 3 · Developer 3

Transactions

The transactions module records money coming in and going out. Each transaction has an amount, a type (expense or income), a category of the same type, a date and an optional description. Amounts are stored in the profile's currency and are always positive; the type gives the direction. Purchases made in another currency are converted at today's rate, and a receipt photo can fill the form.

What it does

- Creates an expense or an income for the active profile.
- Checks that the amount is above zero and that the category matches the transaction type.
- Shows the history, newest first, with category, date, description and a coloured amount.
- Searches descriptions and filters by type, category and date range; filters combine.
- Converts an amount paid in EUR, USD, GBP or MAD into the profile's currency, showing the rate, and keeps the original amount in the description.
- Scans a receipt photo and fills in the total, the date and the shop name. Cash given and change are ignored. The user picks the category and checks before saving.
- Edits or deletes a transaction; budget progress is recalculated, nothing else is removed.

Free APIs used

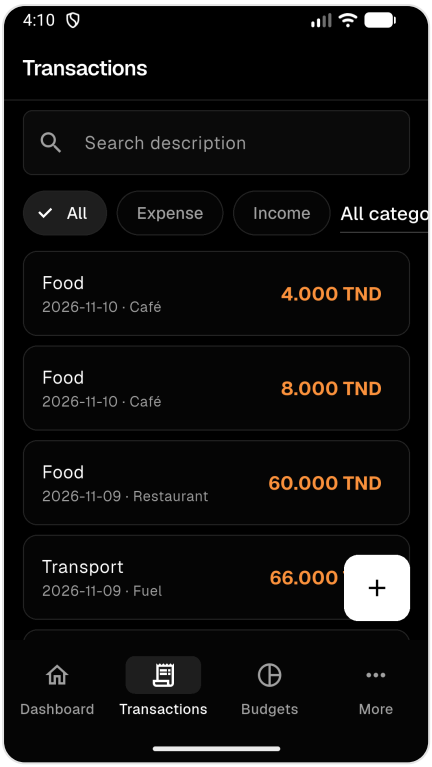
ExchangeRate-API open.er-api.com

Daily exchange rates for the “Paid in” conversion, including TND. The rate is cached for 12 hours and credited on screen as its terms require.

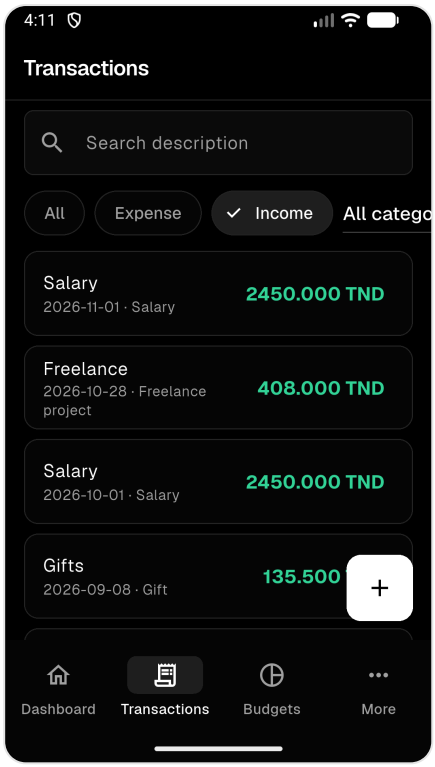
Google ML Kit text recognition on the device, no server

Reads the text of a receipt photo for the scanner. It runs on the phone, works offline and the photo is never uploaded.

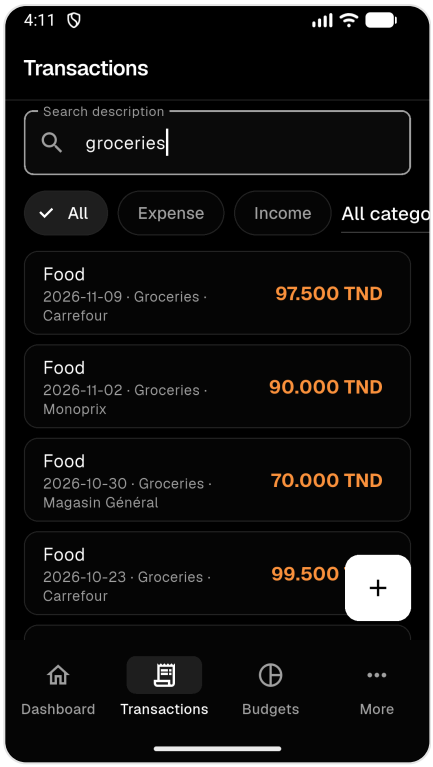
Screens



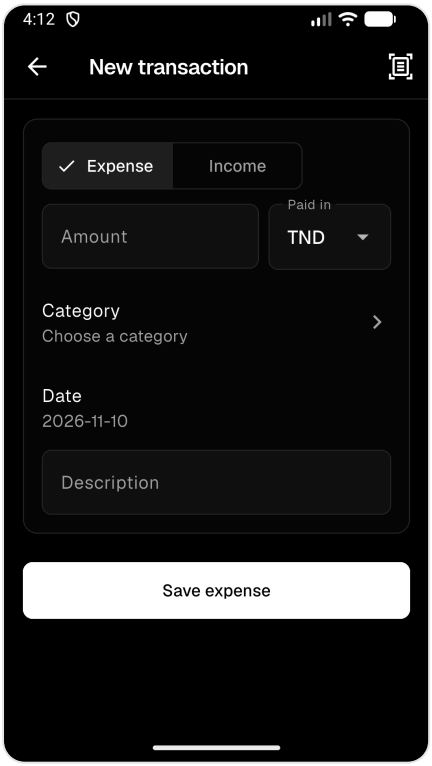
History
All transactions, newest first. Income in green, expenses in orange.



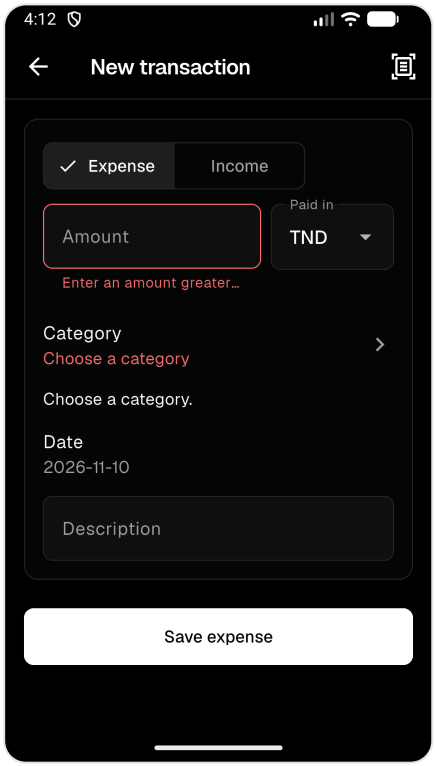
Income filter
Only income transactions.



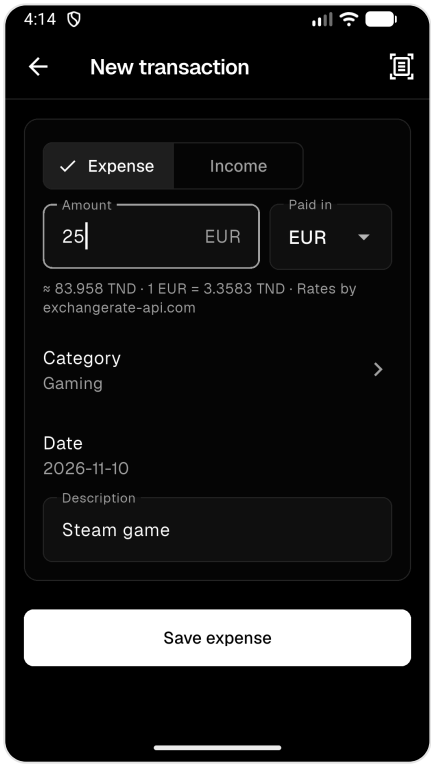
Search
Description search for “groceries”.



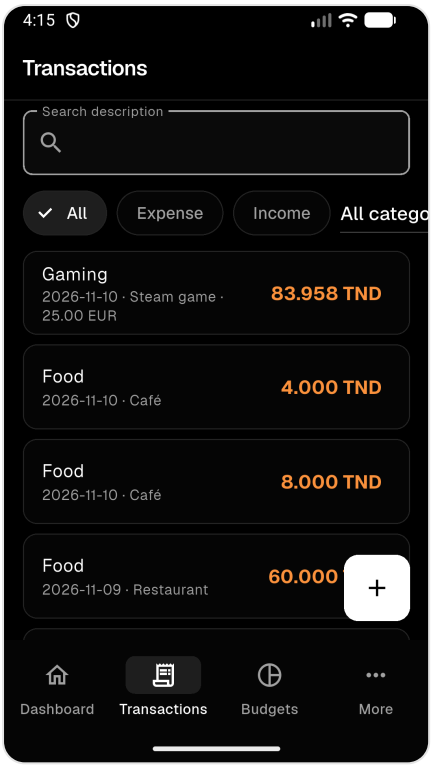
New transaction
Type, amount, “Paid in” currency, category, date and description. The scan button is at the top right.



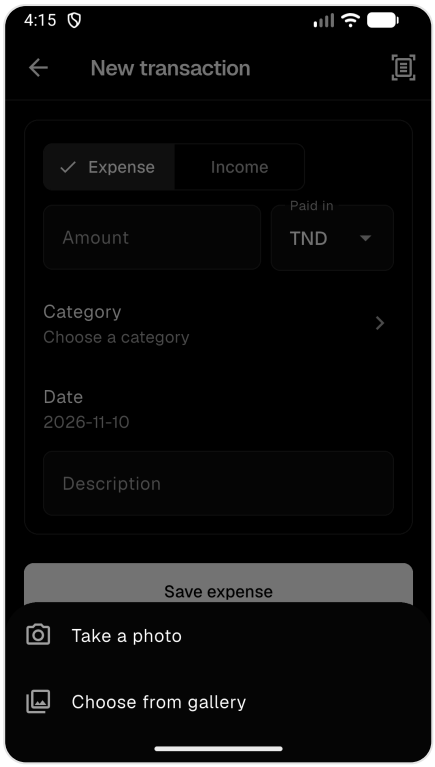
Form validation
A missing amount and category are refused.



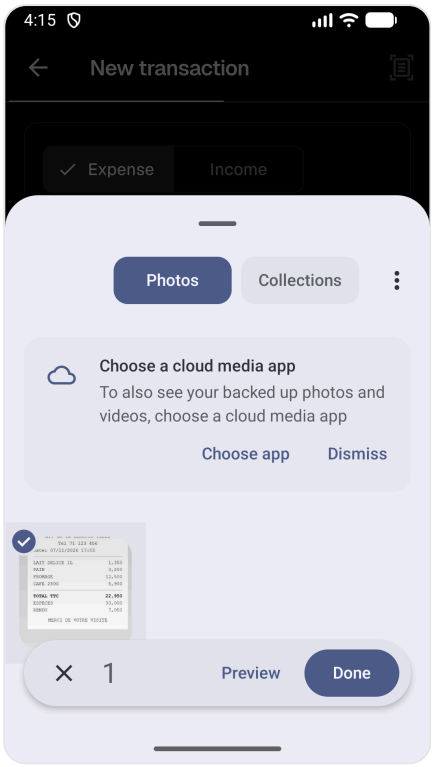
Paid in another currency
25 EUR becomes 83.958 TND at today's rate, with the rate and its source shown.



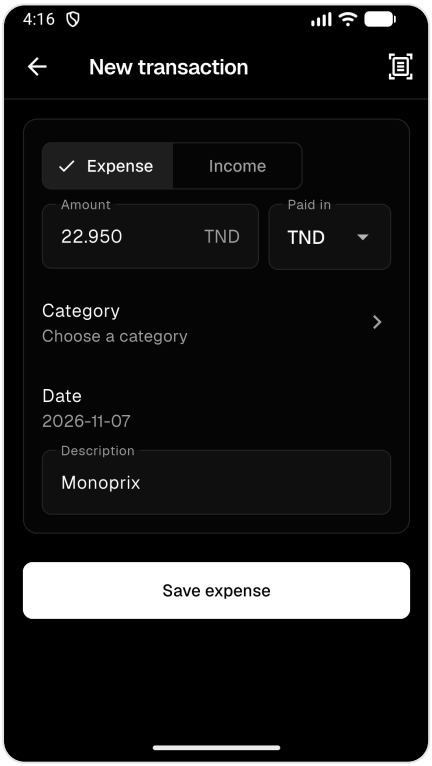
Converted purchase saved
Stored in TND; the original 25.00 EUR stays in the description.



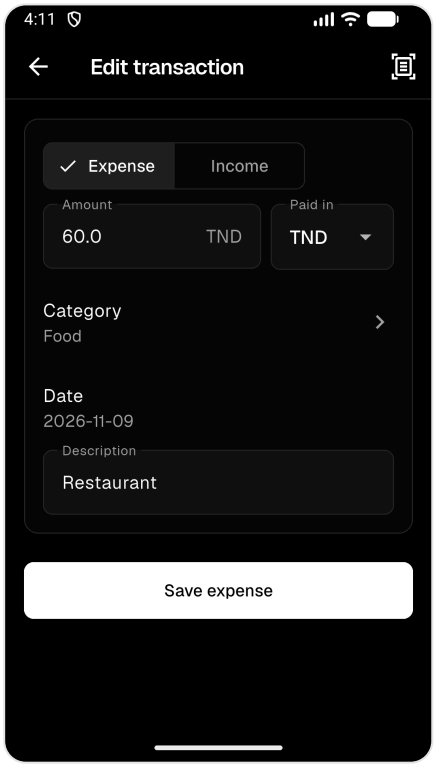
Scan a receipt
Take a photo or choose one from the gallery.



Choosing the photo
The Android photo picker; the app only sees the selected photo.



Filled from the receipt
Total 22.950, date 2026-11-07 and shop Monoprix read from the photo.



Edit transaction
An existing restaurant expense opened for editing.

Module 4 · Developer 4

Budgets

The budgets module sets spending limits. A budget covers one expense category over a period, either a calendar month or custom dates. Spending is never stored on the budget: it is calculated from the expense transactions in that period, so progress is always up to date. Each budget also warns about holidays and Ramadan in its period, when spending usually changes.

What it does

- Creates a budget for an expense category with a positive limit.
- Uses a monthly period (first to last day of a month) or custom start and end dates.
- Refuses income categories and overlapping budgets for the same category.
- Shows the amount spent, what remains and a progress bar for each budget.
- Marks each budget normal, warning (80% used and above) or exceeded (over 100%).
- Lists Tunisian public holidays, Ramadan and the Islamic holidays that fall inside the budget period.
- Edits or deletes a budget. Deleting a budget keeps its transactions.

Free APIs used

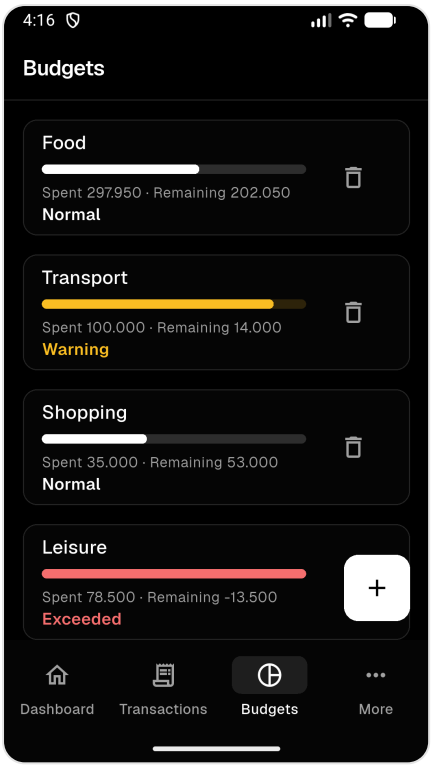
Nager.Date date.nager.at

Tunisian public holidays such as Independence Day and Republic Day.

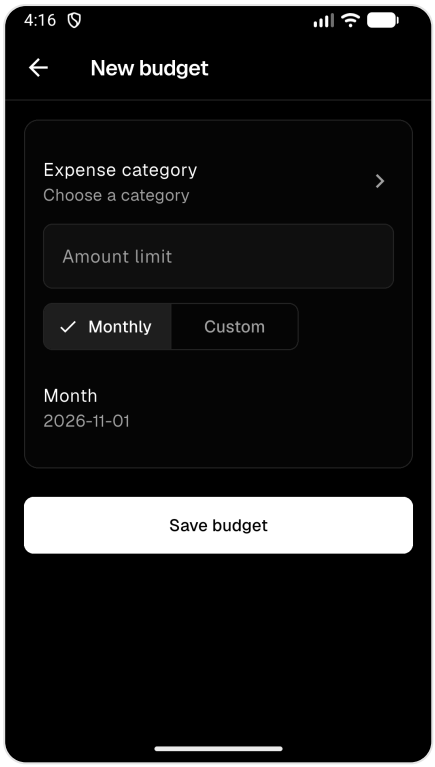
AlAdhan api.aladhan.com

The Hijri calendar, used to find Ramadan, Eid al-Fitr, Eid al-Adha, the Islamic New Year and Mawlid. Both calendar answers are cached for a month.

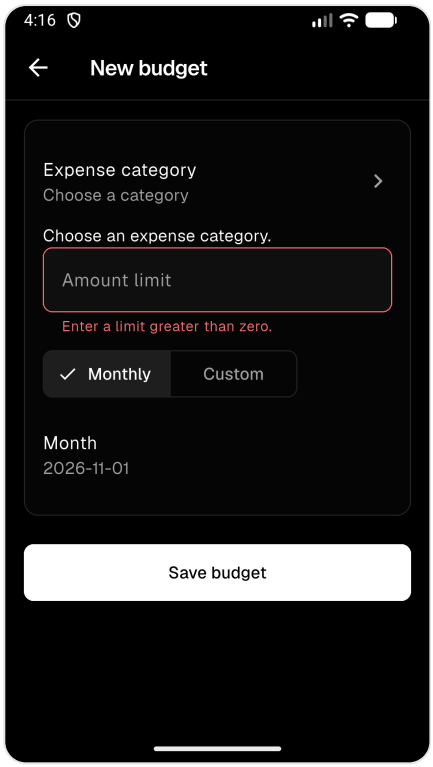
Screens



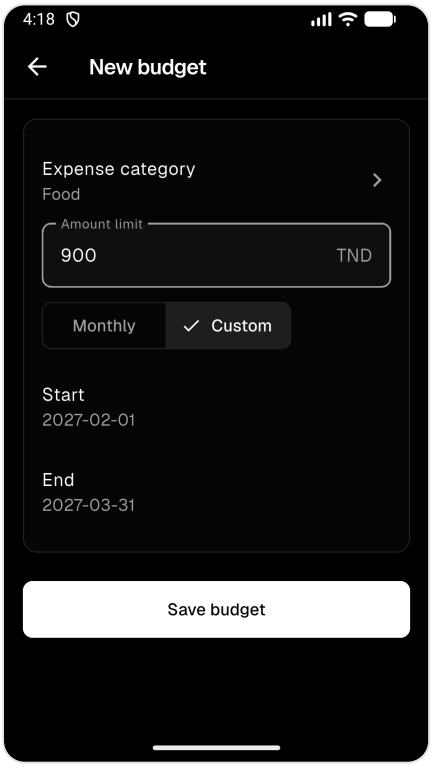
Budgets
This month's budgets in all three states: normal, warning and exceeded.



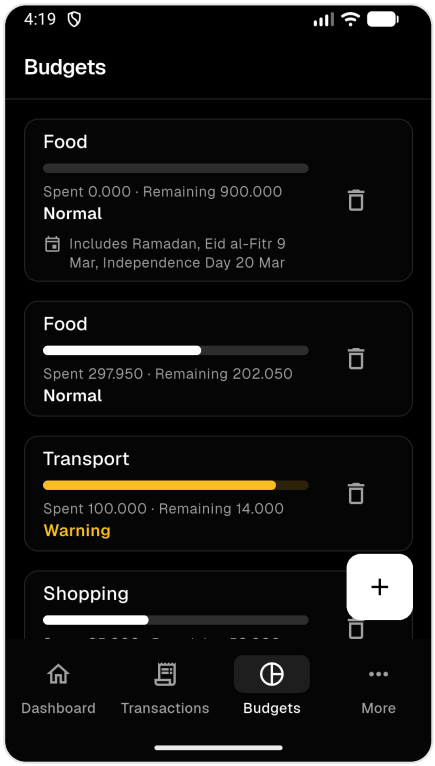
New budget
Expense category, limit and a monthly or custom period.



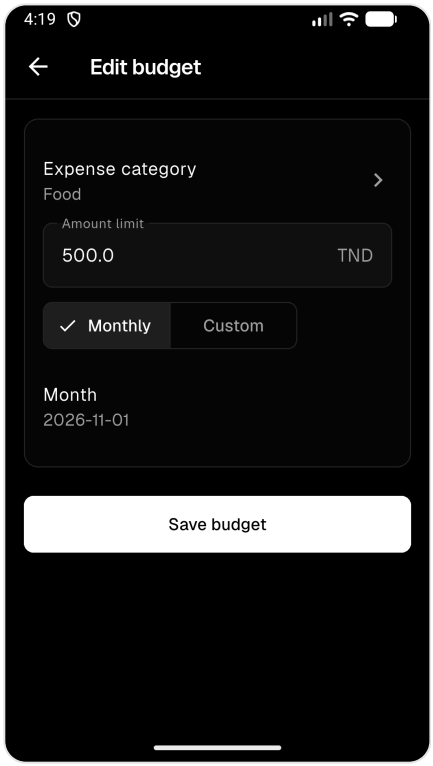
Form validation
A category and a positive limit are required.



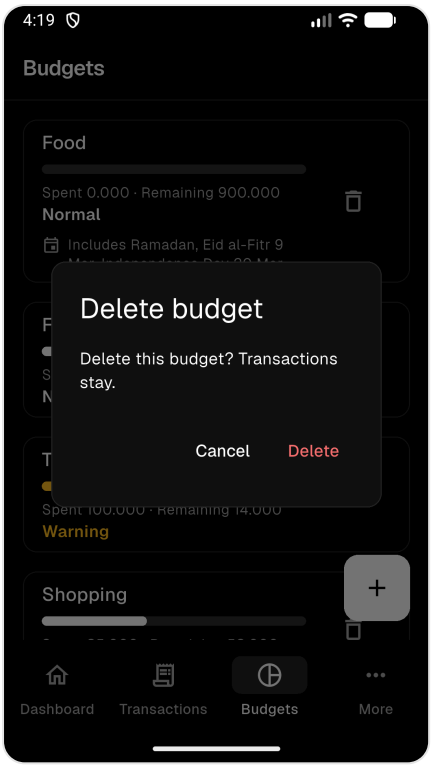
Custom period
A 900 TND Food budget from 1 February to 31 March 2027.



Holidays in the period
The new budget lists Ramadan, Eid al-Fitr and Independence Day.



Edit budget
This month's Food budget opened for editing.



Delete budget

Deleting asks first and keeps the transactions.

Shared · All modules

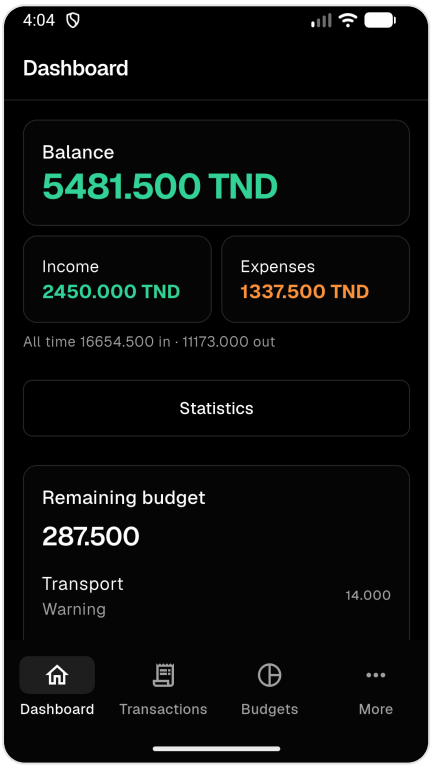
Dashboard and statistics

The dashboard is shared by all four modules. It reads profiles, categories, transactions and budgets without changing them, and is the first screen after choosing a profile.

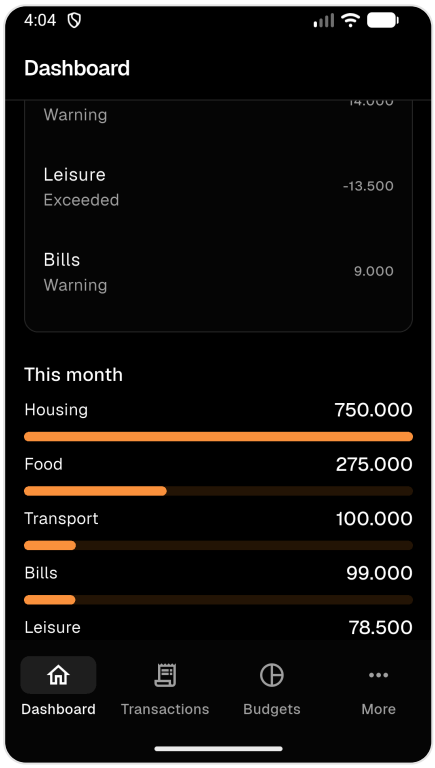
What it does

- Shows the balance, this month's income and expenses, and all-time totals.
- Shows the remaining budget and each budget's state.
- Shows this month's spending by category and the most recent transactions.
- Statistics compares income and expenses month by month and ranks expense categories.

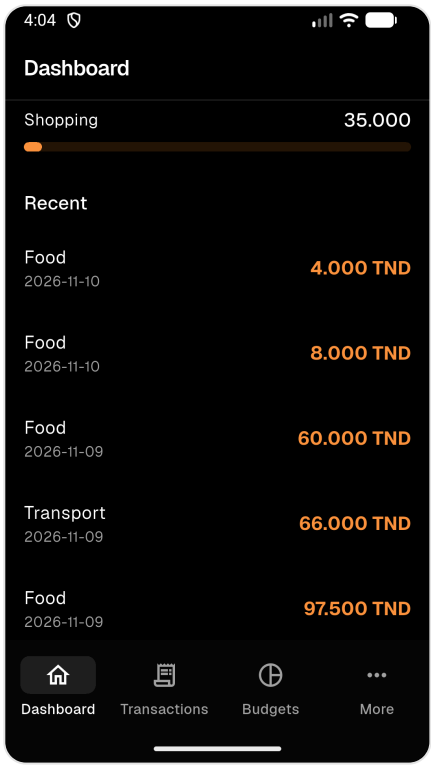
Screens



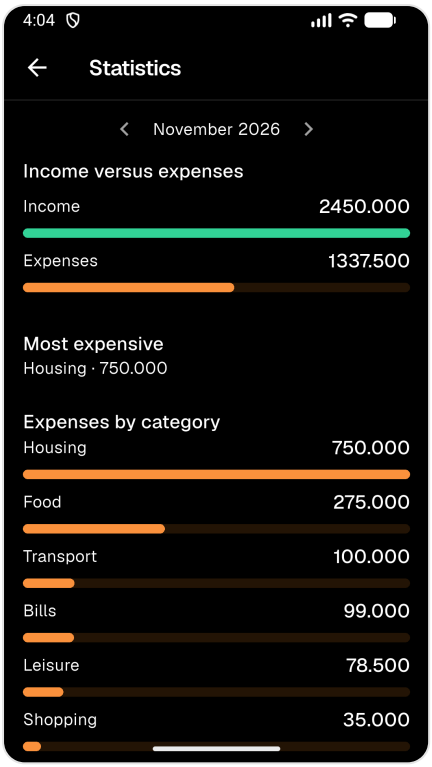
Dashboard
Balance, this month's income and expenses, and the remaining budget.



This month
Budget states and spending by category.



Recent
The latest transactions.



Statistics
Income versus expenses and expenses by category for November 2026.

Screens captured on an Android emulator from the app in its dark theme (one in light), using the built-in demo data, November 2026.